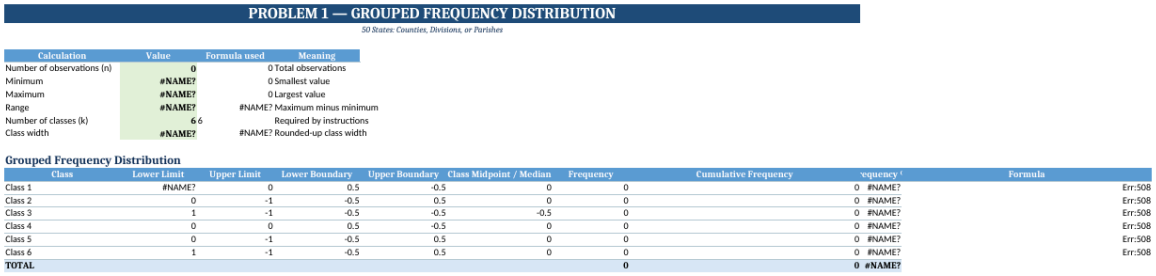
ITC C503 DATA ANALYTICS MIDTERM PROJECT

SCREENSHOT DOCUMENT

Prepared By	Angelo Veno Udrih M. Macalino
Section	BSIT- 303I
Part I	Grouped Frequency Distribution and Statistical Charts
Part II	PivotTable and PivotChart Summaries

PROBLEM 1 — GROUPED FREQUENCY DISTRIBUTION

The screenshot below shows the grouped frequency table with lower limits, upper limits, class boundaries, class midpoint/median, frequency, cumulative frequency, and frequency percentage.



PROBLEM 1 — GROUPED FREQUENCY DISTRIBUTION

50 States: Counties, Divisions, or Parishes

Calculation	Value	Formula used	Meaning
Number of observations (n)	0	0	Total observations
Minimum	#NAME?	0	Smallest value
Maximum	#NAME?	0	Largest value
Range	#NAME?	#NAME?	Maximum minus minimum
Number of classes (k)	6	6	Required by instructions
Class width	#NAME?	#NAME?	Rounded-up class width

Grouped Frequency Distribution

Class	Lower Limit	Upper Limit	Lower Boundary	Upper Boundary	Class Midpoint / Median	Frequency	Cumulative Frequency	'equency ¹	Formula
Class 1	#NAME?	0	0.5	-0.5	0	0	0	#NAME?	Err:508
Class 2	0	-1	-0.5	0.5	0	0	0	#NAME?	Err:508
Class 3	1	-1	-0.5	-0.5	-0.5	0	0	#NAME?	Err:508
Class 4	0	0	0.5	-0.5	0	0	0	#NAME?	Err:508
Class 5	0	-1	-0.5	0.5	0	0	0	#NAME?	Err:508
Class 6	1	-1	-0.5	0.5	0	0	0	#NAME?	Err:508
TOTAL						0	0	#NAME?	

Spreadsheet screenshot — Problem 1

Formula / Computation	Purpose / Result
n = COUNT(data)	50 observations
Minimum = MIN(data)	3
Maximum = MAX(data)	254
Range = Maximum − Minimum	251
Number of classes	6
Class width = ROUNDUP(Range / 6, 0)	42
Lower limit → next class	Previous upper limit + 1
Upper limit	Lower limit + class width − 1
Lower boundary	Lower limit − 0.5
Upper boundary	Upper limit + 0.5
Class midpoint / median	(Lower Limit + Upper Limit) / 2

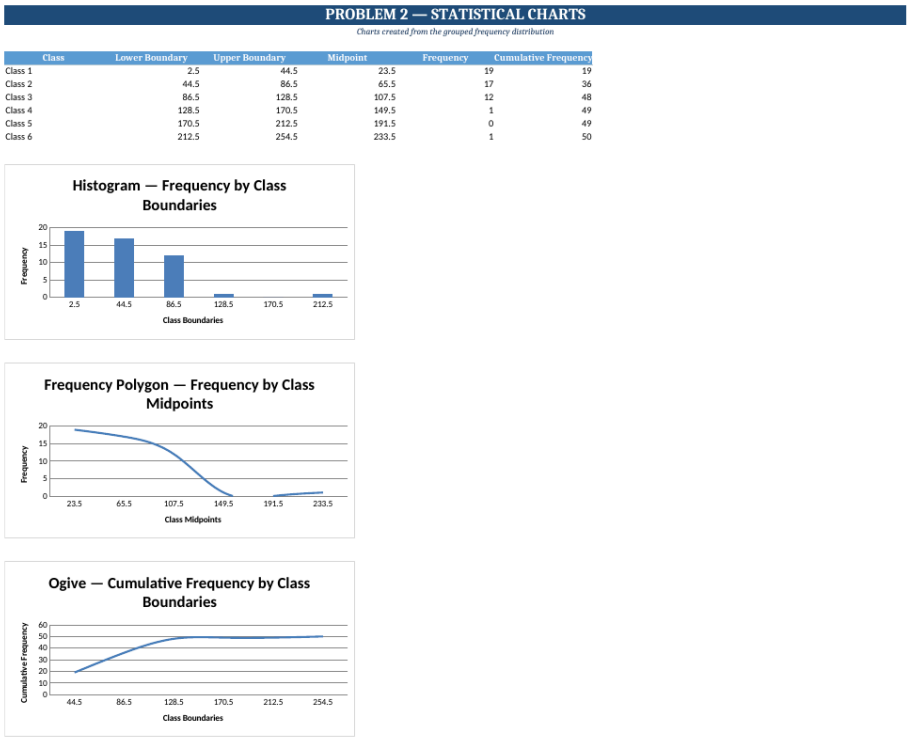
Prepared By: Angelo Veno Udrih M. Macalino

Frequency	COUNTIFS(data, ">="&Lower Limit, data, "<="&Upper Limit)
Cumulative frequency	Running total of frequency
Frequency %	Frequency / 50

Final checks: Total Frequency = 50; Total Frequency Percentage = 100%.

PROBLEM 2 — STATISTICAL CHARTS

The three charts below are based on the completed grouped frequency distribution, as required: Histogram, Frequency Polygon, and Ogive.

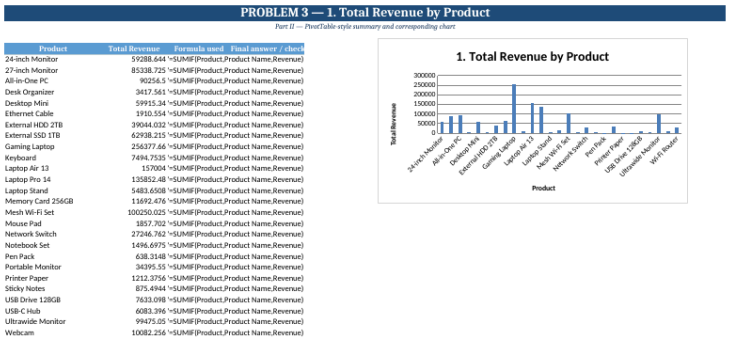


Spreadsheet screenshot — Histogram, Frequency Polygon, and Ogive

Formula / Computation	Purpose / Result
Histogram	Y-axis = Frequency; X-axis = Class Boundaries
Frequency Polygon	Y-axis = Frequency; X-axis = Class Midpoints
Ogive	Y-axis = Cumulative Frequency; X-axis = Class Boundaries
Chart requirements	Titles, axis titles, series names, and data labels are included

PROBLEM 3 — TOTAL REVENUE BY PRODUCT

Part II spreadsheet summary and corresponding chart.



PROBLEM 3 — 1. Total Revenue by Product

Part II — PivotTable-style summary and corresponding chart

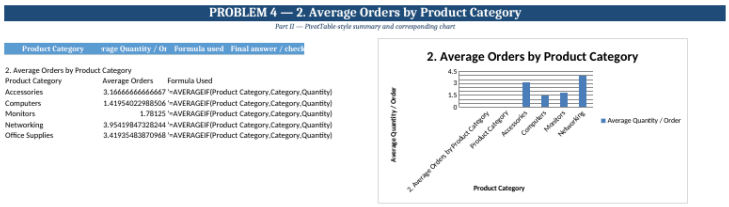
Product	Total Revenue	Formula used	Final answer / check
24-inch Monitor	59288.644	=SUMIF(Product,Product Name,Revenue)	
27-inch Monitor	85338.725	=SUMIF(Product,Product Name,Revenue)	
All-in-One PC	90256.5	=SUMIF(Product,Product Name,Revenue)	
Desk Organizer	3417.561	=SUMIF(Product,Product Name,Revenue)	
Desktop Mini	59915.34	=SUMIF(Product,Product Name,Revenue)	
Ethernet Cable	1910.554	=SUMIF(Product,Product Name,Revenue)	
External HDD 2TB	39044.032	=SUMIF(Product,Product Name,Revenue)	
External SSD 1TB	62938.215	=SUMIF(Product,Product Name,Revenue)	
Gaming Laptop	256377.66	=SUMIF(Product,Product Name,Revenue)	
Keyboard	7494.7535	=SUMIF(Product,Product Name,Revenue)	
Laptop Air 13	157004	=SUMIF(Product,Product Name,Revenue)	
Laptop Pro 14	135852.48	=SUMIF(Product,Product Name,Revenue)	
Laptop Stand	5483.6508	=SUMIF(Product,Product Name,Revenue)	
Memory Card 256GB	11692.476	=SUMIF(Product,Product Name,Revenue)	
Mesh Wi-Fi Set	100250.025	=SUMIF(Product,Product Name,Revenue)	
Mouse Pad	1857.702	=SUMIF(Product,Product Name,Revenue)	
Network Switch	27346.762	=SUMIF(Product,Product Name,Revenue)	
Notebook Set	1496.6975	=SUMIF(Product,Product Name,Revenue)	
Pen Pack	638.3148	=SUMIF(Product,Product Name,Revenue)	
Portable Monitor	34395.55	=SUMIF(Product,Product Name,Revenue)	
Printer Paper	1212.3756	=SUMIF(Product,Product Name,Revenue)	
Sticky Notes	875.4944	=SUMIF(Product,Product Name,Revenue)	
USB Drive 128GB	7633.098	=SUMIF(Product,Product Name,Revenue)	
USB-C Hub	6083.396	=SUMIF(Product,Product Name,Revenue)	
Ultrawide Monitor	99475.05	=SUMIF(Product,Product Name,Revenue)	
Webcam	10082.256	=SUMIF(Product,Product Name,Revenue)	

Spreadsheet screenshot — PROBLEM 3 — TOTAL REVENUE BY PRODUCT

Formula / Computation	Purpose / Result
Formula	SUMIF(Product, Product Name, Revenue)
Final result	Revenue total is calculated separately for each product.

PROBLEM 4 — AVERAGE ORDERS BY PRODUCT CATEGORY

Part II spreadsheet summary and corresponding chart.



PROBLEM 4 — 2. Average Orders by Product Category

Part II — PivotTable-style summary and corresponding chart

Product Category	rage Quantity / Or	Formula used	Final answer / check

2. Average Orders by Product Category

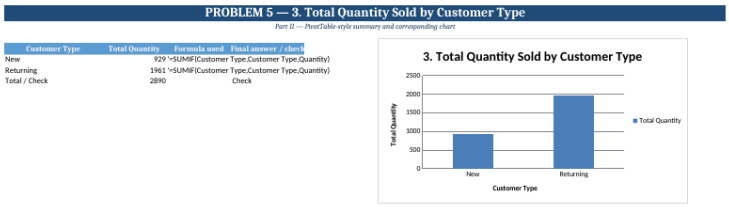
Product Category	Average Orders	Formula Used
Accessories	3.16666666666667	=AVERAGEIF(Product Category,Category,Quantity)
Computers	1.41954022988506	=AVERAGEIF(Product Category,Category,Quantity)
Monitors	1.78125	=AVERAGEIF(Product Category,Category,Quantity)
Networking	3.95419847328244	=AVERAGEIF(Product Category,Category,Quantity)
Office Supplies	3.41935483870968	=AVERAGEIF(Product Category,Category,Quantity)

Spreadsheet screenshot — PROBLEM 4 — AVERAGE ORDERS BY PRODUCT CATEGORY

Formula / Computation	Purpose / Result
Formula	AVERAGEIF(Product Category, Category, Quantity)
Final result	Average quantity/order is calculated for each product category.

PROBLEM 5 — TOTAL QUANTITY SOLD BY CUSTOMER TYPE

Part II spreadsheet summary and corresponding chart.

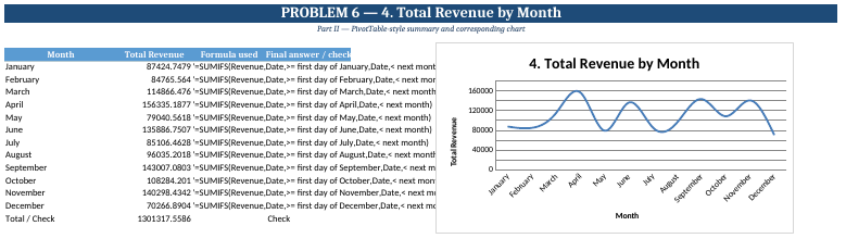


Spreadsheet screenshot — PROBLEM 5 — TOTAL QUANTITY SOLD BY CUSTOMER TYPE

Formula / Computation	Purpose / Result
Formula	SUMIF(Customer Type, Customer Type, Quantity)
Final result	New = 929; Returning = 1,961; Total = 2,890.

PROBLEM 6 — TOTAL REVENUE BY MONTH

Part II spreadsheet summary and corresponding chart.



PROBLEM 6 — 4. Total Revenue by Month

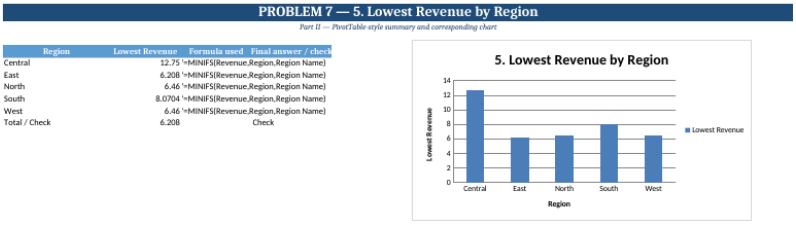
Part II — PivotTable-style summary and corresponding chart

Month	Total Revenue	Formula used	Final answer / check
January	87424.7479	'=SUMIFS(Revenue,Date,>= first day of January,Date,< next mont	
February	84765.564	'=SUMIFS(Revenue,Date,>= first day of February,Date,< next mor	
March	114866.476	'=SUMIFS(Revenue,Date,>= first day of March,Date,< next month	
April	156335.1877	'=SUMIFS(Revenue,Date,>= first day of April,Date,< next month)	
May	79040.5618	'=SUMIFS(Revenue,Date,>= first day of May,Date,< next month)	
June	135886.7507	'=SUMIFS(Revenue,Date,>= first day of June,Date,< next month)	
July	85106.4628	'=SUMIFS(Revenue,Date,>= first day of July,Date,< next month)	
August	96035.2018	'=SUMIFS(Revenue,Date,>= first day of August,Date,< next month	
September	143007.0803	'=SUMIFS(Revenue,Date,>= first day of September,Date,< next m	
October	108284.201	'=SUMIFS(Revenue,Date,>= first day of October,Date,< next mon	
November	140298.4342	'=SUMIFS(Revenue,Date,>= first day of November,Date,< next m	
December	70266.8904	'=SUMIFS(Revenue,Date,>= first day of December,Date,< next m	
Total / Check	1301317.5586		Check

Formula / Computation	Purpose / Result
Formula	SUMIFS(Revenue, Date, first day of month, Date, next month)
Final result	Total annual revenue/check = 1,301,317.5586.

PROBLEM 7 — LOWEST REVENUE BY REGION

Part II spreadsheet summary and corresponding chart.



PROBLEM 7 — 5. Lowest Revenue by Region

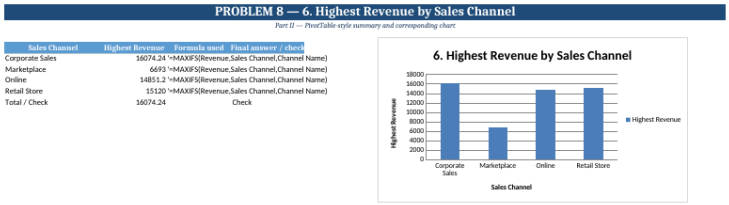
Part II — PivotTable-style summary and corresponding chart

Region	Lowest Revenue	Formula used	Final answer / check
Central	12.75	=MINIFS(Revenue,Region,Region Name)	
East	6.208	=MINIFS(Revenue,Region,Region Name)	
North	6.46	=MINIFS(Revenue,Region,Region Name)	
South	8.0704	=MINIFS(Revenue,Region,Region Name)	
West	6.46	=MINIFS(Revenue,Region,Region Name)	
Total / Check	6.208	Check	

Formula / Computation	Purpose / Result
Formula	MINIFS(Revenue, Region, Region Name)
Final result	Central = 12.75; East = 6.208; North = 6.46; South = 8.0704; West = 6.46.

PROBLEM 8 — HIGHEST REVENUE BY SALES CHANNEL

Part II spreadsheet summary and corresponding chart.



Spreadsheet screenshot — PROBLEM 8 — HIGHEST REVENUE BY SALES CHANNEL

Formula / Computation	Purpose / Result
Formula	MAXIFS(Revenue, Sales Channel, Channel Name)
Final result	Corporate Sales = 16,074.24; Marketplace = 6,693; Online = 14,851.20; Retail Store = 15,120.

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FORMULA GUIDE

FORMULA GUIDE

Actual formulas used in the proje

Item	Formula / Calculation	Purpose
Part I — n	'=COUNT('Part I - Grouped Frequency'!$K$6:$K$55)	Counts the 50 observations.
Part I — minimum	'=MIN('Part I - Grouped Frequency'!$K$6:$K$55)	Smallest county/division/parish count.
Part I — maximum	'=MAX('Part I - Grouped Frequency'!$K$6:$K$55)	Largest county/division/parish count.
Part I — range	'=B8-B7	Maximum minus minimum.
Part I — class width	'=ROUNDUP(B9/B10,0)	Range divided by 6 classes, rounded up.
Part I — lower limit	'=C(previous row)+1	Next class starts one above previous upper limit.
Part I — upper limit	'=B(current row)+$B$11-1	Lower limit plus class width minus one.
Part I — lower boundary	'=B(current row)-0.5	Lower limit minus 0.5.
Part I — upper boundary	'=C(current row)+0.5	Upper limit plus 0.5.
Part I — class midpoint	'=AVERAGE(B(current row),C(current row))	Class midpoint / class median.
Part I — frequency	'=COUNTIFS(data,">="&lower,data,"<="&upper)	Counts observations in the class.
Part I — cumulative	'=SUM(all frequencies through current row)	Running frequency total.
Part I — frequency %	'=Frequency/n	Share of all observations.
Sales_Data — revenue	'=Quantity*Unit Price*(1-Discount)	Calculates revenue per order.
Part II — product revenue	'=SUMIF(Product,Product Name,Revenue)	Adds revenue for each product.
Part II — average orders	'=AVERAGEIF(Product Category,Category,Quantity)	Average quantity per order for each category.
Part II — customer quantity	'=SUMIF(Customer Type,Customer Type,Quantity)	Adds quantity for each customer type.
Part II — monthly revenue	'=SUMIFS(Revenue,Date,month start,Date,next month)	Adds revenue for each month.
Part II — lowest region revenue	'=MINIFS(Revenue,Region,Region Name)	Lowest single-order revenue in each region.
Part II — highest channel revenue	'=MAXIFS(Revenue,Sales Channel,Channel Name)	Highest single-order revenue in each sales channel.

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